Mihir Koladiya

Highway & Transportation Engineer

Road Safety | Crash Analytics | GIS | AI/ML Applications | Blackspot & Hotspot Analysis

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Professional Profile

Highway and Transportation Engineer with a postgraduate background and applied experience in road safety, crash analytics, blackspot investigation, and GIS-based spatial analysis. Skilled in QGIS, ArcGIS, Python, SPSS, and Jamovi for crash data analysis, hotspot mapping, safety trend interpretation, and data-driven road safety assessment. Backed by research experience using AI/ML and geospatial approaches to identify crash-contributing factors, with the aim of advancing transportation infrastructure through sustainable solutions that enhance safety, efficiency, and evidence-based decision-making. Experienced in field surveys, road safety audits, DGPS-based road inventory, highway safety evaluation, and technical report preparation for transportation infrastructure projects.

Core Competencies and Technical Expertise

Core Area	Applications / Expertise	Tools / Platforms
Crash Data Analytics	Crash data cleaning, crash frequency and severity analysis, contributing-factor identification, blackspot diagnosis, safety trend interpretation, and crash-risk assessment.	Python, SPSS, Jamovi, MS Excel
GIS-Based Hotspot Mapping	Crash mapping, geocoding support, hotspot identification, KDE-based heatmaps, spatial clustering, road inventory analysis, and corridor-level safety interpretation.	QGIS, ArcGIS, Google Earth Engine
Road Safety Audit & Field Validation	Road safety audits, field inspections, roadway geometry assessment, signage and marking evaluation, risk diagnosis, blackspot inspection, and safety countermeasure planning.	IRC Guidelines, AutoCAD, MS Excel
Statistical & Crash-Risk Modelling	Poisson regression, Negative Binomial regression, Zero-Inflated models, Hurdle models, GWR, model comparison, and result interpretation.	SPSS, Jamovi, R, Python
AI/ML for Road Safety	Classification, regression, feature importance analysis, predictive crash modelling, Random Forest, XGBoost, ANN, and AI/ML-based safety diagnostics.	Python, XGBoost, Random Forest, ANN
Transport Planning & Mobility Analysis	Mode choice analysis, accessibility assessment, travel behaviour analysis, sustainable mobility evaluation, low-carbon mobility planning, and policy-oriented interpretation.	R, Python, SPSS, SmartPLS, MS Excel
Highway Geometric Design & Safety	Roadway geometry assessment, intersection layout review, traffic safety evaluation, pavement-related interpretation, field-data analysis, and design-based safety recommendations.	AutoCAD, IITPave, MS Excel
Technical Reporting & Documentation	Research paper writing, technical report preparation, road safety audit documentation, stakeholder presentations, project documentation, and academic presentation development.	MS Word, PowerPoint, LaTeX

Education

Degree	Branch / Specialization	Institute	CGPA/%	Year
M.E.	Highway & Transportation Engg.	The M.S. University of Baroda, Vadodara	8.8/10	2023–2025
B.E.	Civil Engineering	SSASIT, Surat	8.92/10	2019–2022
D.E.	Civil Engineering	Tapi Diploma Engineering College, Surat	8.22/10	2016–2019
10th	Secondary School Certificate	Vivekanand Vidyasankul, Amreli	67.33%	2016

Experience

Professional Experience

Project Engineer – Transportation Engineering and Planning *May 2026 – Present*
Sardar Vallabhbhai National Institute of Technology (SVNIT) | Surat, India

- Working on transportation engineering and planning activities under a highway safety and risk analytics project.
- Analysing crash data for spatiotemporal trends, hotspot detection, and proactive road safety assessment.
- Supporting highway safety analysis through GIS, statistical methods, and data-driven analytical techniques.
- Conducting transportation data processing, spatial analysis, visualization, and preparation of technical project outputs.
- Contributing to research, technical reporting, and development of data-driven approaches for highway safety and risk assessment.

Teaching Experience

Lecturer – Civil Engineering *Aug 2025 – Mar 2026*
Sigma University | Vadodara, India

- Taught undergraduate courses in traffic, transportation, surveying, AutoCAD, and infrastructure planning.
- Conducted practical sessions, assignments, evaluations, and software-based learning for civil engineering students.
- Guided student projects on road safety, traffic studies, pavement design, GIS analysis, and infrastructure planning.
- Prepared lecture notes, technical presentations, laboratory content, and assessment material for academic delivery.
- Mentored students in field-oriented problem solving, technical report writing, and project-based learning.

Academic Leadership

Department Head Coordinator – PG Civil Engineering, Training and Placement Cell *Mar 2024 – May 2025*
Faculty of Technology and Engineering, MSU Baroda | Vadodara, India

- Coordinated placement communication among PG students, faculty members, recruiters, and department leadership.
- Managed student profiles, resume screening, recruiter follow-ups, interview schedules, and placement records.
- Organized company visits, technical sessions, and soft-skill activities to support career readiness.
- Assisted students with profile review, professional communication, and preparation for recruitment processes.
- Supported departmental reporting, placement documentation, and industry-academia coordination activities.

Road Safety Internship Experience

Road Safety Engineer Intern *May 2024 – Jun 2024*
Mattest Engineering Services | Surat, India

- Conducted rural road safety audits to identify blackspots, high-risk locations, and geometric deficiencies.
- Assessed roadway geometry, sight distance, visibility, signage, markings, shoulders, and junction layouts.
- Documented site observations using field notes, photographs, safety issues, and location-specific deficiencies.
- Applied IRC:SP-88 and IRC:SP-84 guidelines for safety diagnosis and countermeasure planning.
- Prepared audit findings and crash mitigation suggestions for field-based road safety improvement.

Field Experience

Junior Civil Engineer *May 2022 – Jul 2023*
TD Patel & Company | Surat, India

- Supervised roadway infrastructure works including layout checking, site inspection, and execution monitoring.
- Verified construction materials, workmanship quality, field measurements, and site-level work progress.
- Prepared quantity records, billing documents, daily progress updates, measurement sheets, and site reports.
- Coordinated with engineers, vendors, labour teams, and clients for smooth execution of construction activities.
- Supported quality control, resource planning, site issue resolution, and implementation-level engineering decisions.

Internship Experience

Civil Engineer Intern *Jan 2022 – Apr 2022*
Mattest Engineering Services | Surat, India

- Conducted DGPS-based road inventory and topographic surveys across 300+ km of highway corridors.
- Collected roadway features, geometric details, pavement observations, chainage references, and field records.
- Supported highway asset documentation, field validation, pavement condition review, and inventory preparation.
- Assisted in bituminous mix design compliance, pavement QA/QC records, and material testing documentation.
- Compiled survey data and technical records for project planning, validation reporting, and quality assessment.

Civil Engineer Intern *Sep 2021 – Oct 2021*
Pramukh Test House & Consultancy Services | Surat, India

- Performed laboratory testing of soil samples for pavement subgrade evaluation and material assessment.
- Conducted bitumen testing to support pavement material evaluation and highway quality control.
- Prepared sample records, test observations, result summaries, and laboratory documentation.

- Assisted in QA/QC documentation for soil, bitumen, and pavement-related material tests.
- Supported technical reporting and data compilation for highway material quality verification.

Civil Engineer Intern *May 2021 – Aug 2021*

KKB Projects Pvt. Ltd. | Surat, India

- Monitored highway construction activities including subgrade, GSB, WMM, and bituminous layer execution.
- Observed compaction, layer thickness, material placement, and pavement construction quality.
- Assisted in field measurements, execution tracking, material observations, and site documentation.
- Supported QA/QC records, daily progress tracking, and construction activity documentation.
- Assisted site engineers in work verification, site coordination, and pavement layer execution monitoring.

Research and Technical Projects

M.Tech. Dissertation

Exploring the Root Causes of Traffic Crashes Using AI/ML and Geospatial Approaches *2025*

The Maharaja Sayajirao University of Baroda, Vadodara

Supervisor: Dr. Pankaj S. Prajapati, Associate Professor, Department of Civil Engineering, MSU Baroda
Methods and Tools: ArcGIS, QGIS, Python, KDE, Poisson Regression, Negative Binomial Model, Zero-Inflated Model, Hurdle Model, XGBoost.

- Investigated crash factors by analyzing Land Use, Built Environment, population density and road geometry.
- Analyzed crash frequency and severity along NH-48 and NE-01 using statistical, geospatial, and AI/ML modeling.
- Identified spatial crash clusters and safety determinants using GIS-based mapping and predictive modeling.
- Formulated data-driven recommendations for crash mitigation, blackspot treatment, and road safety improvements.

Road Safety Project

Detailed Investigation, Analysis, and Mitigation Measures for Blackspot Improvement *2024*

Client: Gujarat Road Safety Authority

Advisor: Dr. Pankaj S. Prajapati, Associate Professor, MSU Baroda

Project Focus: Blackspot analysis, crash interpretation, field assessment, and safety countermeasures.

- Pinpointed high-risk locations through crash data review, traffic analysis, and comprehensive field diagnostics.
- Evaluated roadway geometry, intersections, visibility, signage, and infrastructure deficiencies for safety compliance.
- Proposed IRC-guided mitigation measures, including geometric improvements, traffic calming, and advanced signage.
- Delivered actionable technical recommendations for stakeholder review, safety audits, and field implementation.

Publications and Scholarly Contributions

Conference Papers Published

1. **Mihir Koladiya**, Khushi Mahajan, and Pankaj S. Prajapati. "Impact of Mobile Use and Travel Behavior on Perceived Crash Risk Among Two-Wheeler Riders." *15th International Conference on Transportation Planning and Implementation Methodologies for Developing Countries (TPMDC)*, IIT Bombay, India, 2024. DOI: 10.1007/978-981-96-8118-1_6.
2. Pankaj S. Prajapati and **Mihir Koladiya**. "Developing Low-Carbon Mobility Strategies: A Roadmap for the City of Vadodara." *International Conference on Environmental Science and Technology*, SVNIT Surat, India, 2024. DOI: 10.1007/978-981-95-1442-7_26.
3. **Mihir Koladiya**, Pankaj S. Prajapati, and Rajankumar J. Patel. "Parking Adequacy and Commuter Behavior at Ahmedabad Metro Stations: An Observational Analysis." *11th Conference on Transportation Systems Engineering and Management (CTSEM)*, NIT Bhopal, India, 2025. DOI: 10.1007/978-981-95-9063-6_19.

Conference Papers Under Publication Process

1. Pankaj S. Prajapati, **Mihir Koladiya**, and Mohammad Huzaif Nizamuddin Khan. "Mode Choice Analysis Based on Mode-Specific Utilities for Work Trips Using Multinomial Logit Model Approach." *3rd International Conference on Transportation Infrastructure Projects: Conception to Execution (TIPCE)*, IIT Roorkee, India, 2025.
2. **Mihir Koladiya** and Pankaj S. Prajapati. "Evaluating Road Traffic Noise Pollution and Its Health Impacts on Residents Near Highways." *3rd International Conference on Transportation Infrastructure Projects: Conception to Execution (TIPCE)*, IIT Roorkee, India, 2025.
3. Khushi Mahajan, Pankaj S. Prajapati, and **Mihir Koladiya**. "Land Use–Transport Integration Through Users' Travel Behaviour Perspective: A Case of Vadodara City." *3rd International Conference on Transportation Infrastructure Projects: Conception to Execution (TIPCE)*, IIT Roorkee, India, 2025.
4. Krunal Kanzariya, **Mihir Koladiya**, Khushi Mahajan, and Pankaj S. Prajapati. "Modeling Private Transportation Mode Selection in Developing Cities Using Multinomial Logistic Regression." *11th Conference on Transportation Systems Engineering and Management (CTSEM)*, NIT Bhopal, India, 2025.

Conference Papers Presented

1. Sahil Vora, Dhara Prajapati, **Mihir Koladiya**, and Pankaj S. Prajapati. "Factors Influencing Traffic Crashes in Vadodara: A Comparative Study of City Roads and National Highways Using Artificial Neural Networks." *3rd International Conference on Transportation Infrastructure Projects: Conception to Execution (TIPCE)*, IIT Roorkee, India, 2025.

Peer Review Contribution

1. Reviewed manuscript titled "Smart and Sustainable Technologies in Present and Future Parking Structures" for the *Journal of Engineering Research and Reviews*, 12 Apr 2025.

Certifications, Workshops and Awards

Certifications

Date	Title	Organized By	Mode
18–21 Sep 2025	Paper Presentation Certificate – TIPCE 2025	IIT Roorkee	Offline
04–05 Jul 2025	Paper Presentation Certificate – CTSEM 2025	MANIT Bhopal & CSIR-CRRI	Offline
12 Apr 2025	Certificate of Reviewing – Journal of Engineering Research & Reviews	eJManager	Online
18–20 Dec 2024	Paper Presentation Certificate – TPMDC 2024	IIT Bombay	Offline

Workshops and Training

Date	Title	Organized By	Mode
03–07 Feb 2025	Safe System Perspective on Vulnerable Road Users in India	NIT Warangal	Offline
18 Nov 2024	Advancing Travel Demand Models Using Emerging Data Sources	IIT Bombay	Offline
19–23 Aug 2024	AI/ML for Geodata Analysis	ISRO–IIRS	Online
11–17 Mar 2024	Advanced Methods for Transportation Data Analysis	NIT Warangal	Offline
15 Feb 2020	Workshop on Revit Architecture	SVNIT Surat	Offline
11 Feb 2020	Workshop on AutoCAD	SVNIT Surat	Offline

Awards and Achievements

Date / Year	Achievement	Institution / Event	Outcome
2026	Outstanding Academic Guide Award	Wonder Cement Ltd. / Concrete Catalyst	Award
2026	Mentor Appreciation Certificate	Concrete Catalyst	Appreciation
27 Jun 2025	Appreciation Letter – Road Safety Study	Gujarat Road Safety Authority	Appreciation
01 May 2025	Appreciation Certificate – T&P Cell	The M.S. University of Baroda	Appreciation
24 Apr 2021	Design Engineering Project Fair	SSASIT, Surat	3rd Rank
07–08 Mar 2019	Project/Model Presentation	Tapi Diploma Engineering College, Surat	2nd Rank

Languages

Gujarati: Native proficiency | **Hindi:** Full professional proficiency | **English:** Full professional proficiency

References

Prof. Shriniwas S. Arkatkar
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Relation: Project Principal Investigator

Dr. Pankaj S. Prajapati
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Relation: Dissertation Supervisor / Research Guide